

3 grams of a particular triglyceride (MW: 500 g/mol) are completely saponified under basic conditions to break all ester bonds, and the products of this reaction are then extracted with dilute aqueous acid. The fatty acid products are then dried and quantified via titration. What volume of 1M titrant is needed to reach the equivalence point?

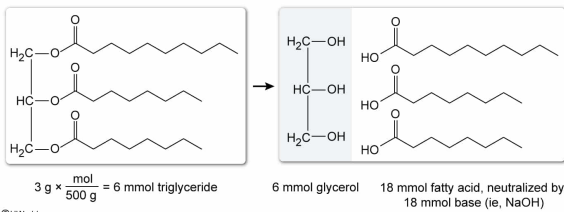
- ☐ A. 2 mL HCl [6%]
☐ B. 6 mL NaOH [43%]
☐ C. 18 mL HCl [21%]
☒ D. 18 mL NaOH [28%]

Correct

28% answered correctly

Explanation:

Each molecule of triglyceride is made of one glycerol and three fatty acids



Triglycerides are **energy-storage lipids** comprising a glycerol backbone that is joined to **three** fatty acids by **ester** linkages. When energy is needed, the **ester linkages** are **hydrolyzed**, releasing free fatty acids and free glycerol. The free fatty acids are transported to the mitochondria and eventually oxidized by **β oxidation**, whereas the glycerol backbone is metabolized into dihydroxyacetone phosphate to enter glycolysis or gluconeogenesis.

In this question, the triglyceride ester linkages are chemically broken through a process called saponification. Despite being broken nonenzymatically, these ester hydrolysis reactions still produce **three fatty acids** and one glycerol molecule. When extracted with **dilute aqueous acid**, the fatty acids remain **protonated**. The fatty acids are dried and the amount of fatty acids can be determined by **titration** using a **base (eg, NaOH)** to deprotonate the fatty acids.

The question states that 3 grams of a 500 g/mol triglyceride molecule are saponified. **Converting to moles** yields

$$3 \text{ g} \times \frac{1 \text{ mol triglyceride}}{500 \text{ g}} = 0.006 \text{ mol} = 6 \text{ mmol triglyceride}$$

Because three fatty acids are released per triglyceride, *three times this amount* (ie, 6 mmol $\times$ 3 = **18 mmol**) of fatty acids are produced. Given that the titrant is 1 M (ie, 1 mol/L), **18 mL of 1M NaOH** is needed to deprotonate the 18 mmol of fatty acids.

(Choice A) 2 mL would be calculated if the amount of triglycerides were *divided* by 3 instead of multiplied by 3. Furthermore, a titration measuring the deprotonation of fatty acids would require a *base* like NaOH, not an *acid* like HCl.

(Choice B) 6 mL of NaOH would be appropriate if only *one* fatty acid were released per triglyceride instead of three.

(Choice C) 18 mL of *NaOH*, not HCl, is required to deprotonate the 18 mmol of fatty acid produced by saponification.

Educational objective:

Triglycerides are composed of a glycerol backbone joined to three fatty acids by ester linkages. Fatty acids are acids and can be quantified by titration with a base.

Time spent:0 sec

Subject:
Biochemistry

QID:403870

System:
Carbohydrates, Nucleotides, and Lipids